



EXAMINATIONS COUNCIL OF ESWATINI
Eswatini General Certificate of Secondary Education

CANDIDATE
NAME

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CENTRE
NUMBER

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CANDIDATE
NUMBER

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BIOLOGY

6884/04

Paper 4 Alternative to Practical Test

October/November 2025

1 hour

Candidates answer on the Question Paper.
Additional materials: Calculator

READ THESE INSTRUCTIONS FIRST

Write your centre number, candidate number and name in the spaces provided.
Write your answers in dark blue or black pen.
You may use an HB pencil for any diagrams, graphs or rough working.
Do **not** use staples, paper clips, glue or correction fluid.
Do **not** write on the bar code.

Answer **all** questions.

You may use an electronic calculator.
You may lose marks if you do not show your working or if you do not use appropriate units.
The number of marks is given in brackets [] at the end of each question or part question.

The total number of marks is 40.

For Examiner's Use	
1	
2	
Total	

This document consists of **8** printed pages.

- 1** When plant tissue is placed in solutions with different water potentials, water molecules move between the cells and the solution.

A student investigated the effect of immersing plant tissue in solutions with different water potentials.

The student was provided with a potato and beakers **A** and **B**, each containing 80 cm^3 of salt solution but with different water potentials.

The student cut two cylinders of potato tissue, **A** and **B**, and placed them on a white tile.

Both potato cylinders were of the **same** length, l .

Fig. 1.1 shows the actual lengths of the potato cylinders.

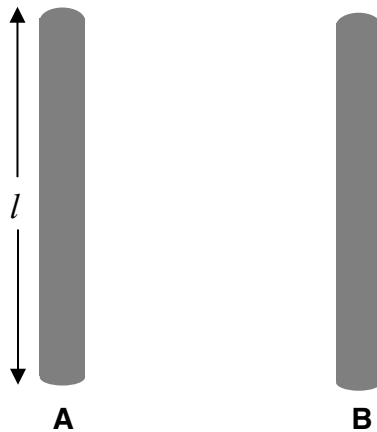


Fig.1.1

- (a) (i)** Measure and record the length of potato cylinders **A** and **B** shown in Fig. 1.1.

length of potato cylinders **A** and **B**. [1]

Potato cylinder **A** was immersed in the solution in beaker **A** and potato cylinder **B** was immersed in the solution in beaker **B**.

The potato cylinders were left in the solutions for 24 hours. They were then removed and had any surface solution dried with a paper towel.

Fig. 1.2 shows the actual potato cylinders after 24 hours.



Fig.1.2

- (ii) Measure and record the lengths of potato cylinders **A** and **B** after 24 hours shown in Fig. 1.2.

A

B

[2]

- (iii) Use your measurements to calculate and record the actual change in length of each potato cylinder.

change in length for potato cylinder **A**

change in length for potato cylinder **B**

[2]

- (b) The student felt the textures of the two potato cylinders after 24 hours.

Suggest what the texture of each potato cylinder may have been.

texture of potato cylinder from solution **A**

.....

texture of potato cylinder from solution **B**

.....

[2]

- (c) Use your measurements and your answer to **1(b)** to explain what happened to bring about changes in the potato cylinder in solution **B**.

.....

.....

.....

.....

.....

..... [3]

- (d) State the dependent variable in this investigation.

..... [1]

- (e) State **two** variables that should have been controlled in this investigation.

1

2 [2]

- (f) State why it would be important to repeat this investigation.

.....

..... [2]

- (g) In a different experiment, some students investigated the effect on similar potato cylinders of immersing them in sucrose solutions of different concentrations.

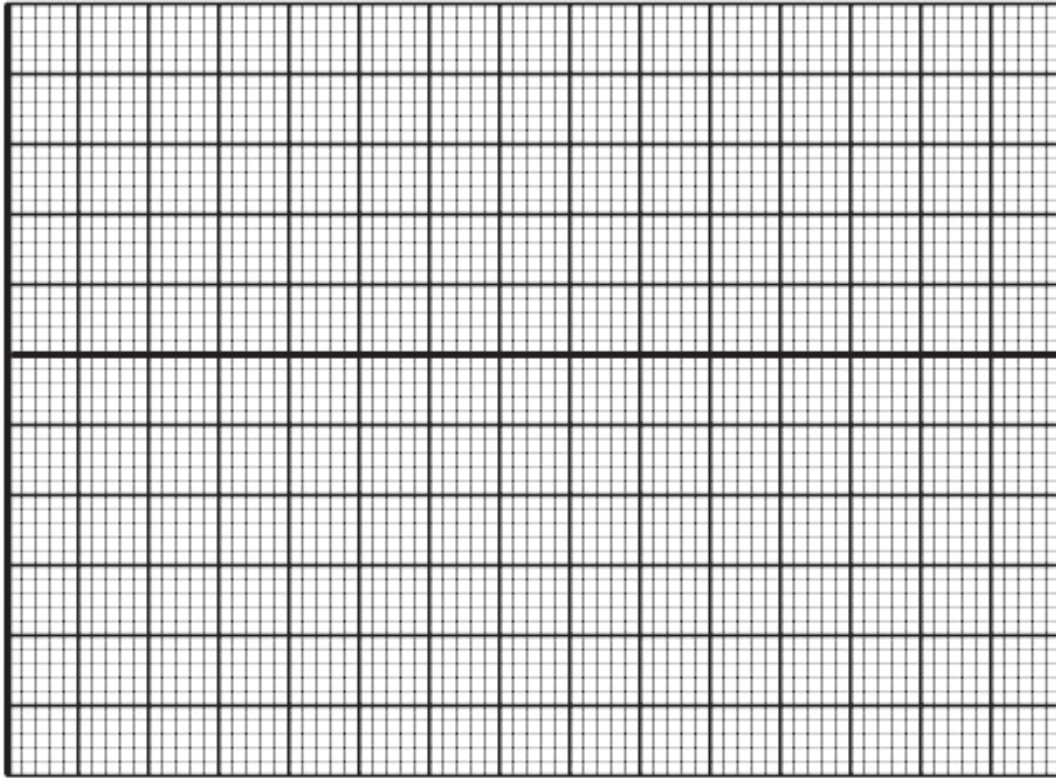
Table 1.1 shows the results of the experiment.

Table 1.1

<u>concentration of sucrose solution</u> mol/dm ³	change in length/mm
0.0	10
0.2	4
0.4	−2
0.6	−7
0.8	−10

- (i) On the grid, plot a line graph of the data shown in Table 1.1.

Join your plotted points with ruled, straight lines.



[5]

- (ii) State the conclusion for this investigation.

.....

..... [1]

- (iii) Use your graph to find the concentration of sucrose in which the length of the potato cylinder did not change. Show your working on your graph.

..... [2]

[Total: 23]

2 Fig. 2.1 shows photographs of the surfaces of two fruits, **C** and **D**, that have both been cut in half.

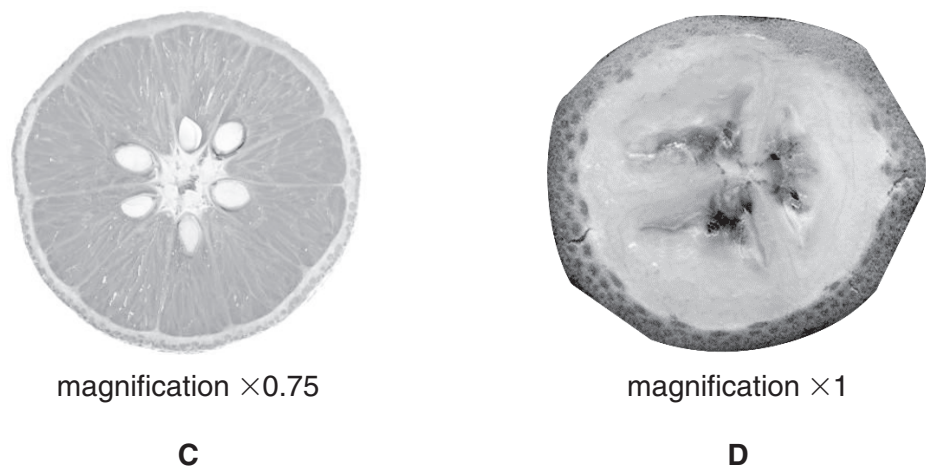


Fig. 2.1

(a) (i) State **one** visible similarity in the fruits **C** and **D**.

.....

..... [1]

(ii) Complete Table 2.1 with **three** visible differences between the fruits.

Table 2.1

	C	D
difference 1		
difference 2		
difference 3		

[3]

- (b) In the space below, make a large drawing of fruit **C** as it appears in Fig. 2.1.

[5]

- (c) (i) Measure and record the length of the diameter of fruit **C** shown in Fig. 2.1.

length of diameter of fruit **C** mm [1]

- (ii) Draw a line across the diameter of your drawing of fruit **C**.

Measure and record the length of the diameter of your drawing of fruit **C**.

length of diameter of your drawing of fruit **C** mm [1]

- (iii) Using your measurements in **2(c)(i)** and **2(c)(ii)**, calculate the magnification of your drawing of fruit **C** compared to the fruit **C** as it appears in Fig. 2.1.

Give your answer to **two** decimal places.

..... [3]

- (d) A student added drops of iodine solution to the cut surfaces of fruits **C** and **D**.

A blue-black colour was seen on fruit **D** but not on fruit **C**.

State what the student could conclude about fruits **C** and **D**.

specimen **C**

specimen **D**

[1]

- (e) The student added some reagent to samples of fruits **C** and **D** and then heated the mixtures, looking for an orange precipitate to be formed.

State the name of the reagent that was added and what the orange precipitate would show.

name of reagent

what the orange precipitate would show

..... [2]

[Total:17]